

The most important commands in seismic handler (all in SH>)

- Opening a window, closing, help
 - **xopen;;** open graphics window
 - **help** displays all available commands
 - **help <command>** displays the explanation for a command
 - **quit** quit SHC

- Read and Write data
 - **read <file> all** read all traces of q files named <file>
 - **write <file> all** save q file named <file>
 - **write<file> 3-6** save q files only with traces 3 to 6

- Displaying the information of the event
 - **list_all origin** origin time: start time of the earthquake
 - **list_all start** start time: start time of the trace
 - **list_all lat** lat: latitude of the epicenter
 - **list_all lon** lon: longitude of the epicenter
 - **list_all depth** depth: depth of the hypocenter in km

- Set header
 - **Set/file all origin <date and time>** set origin time, 23-JAN-2021_01:02:45
 - **Set/file all depth <depth>** set depth in km
 - **Epi_dist <lat> <lon>** set latitude and longitude of the epicenter

- Redrawing, zooming, hiding, deleting
 - **rd** redraw window
 - **yinfo distance** rearrange according to distance
 - **yinfo none** rearrange according to number (back to normal display of the traces)

 - **stw** setting time window using mouse
 - **stw <start time> <end time>** setting the time window using start and end time

 - **dtw** delete time window
 - **cut all <start time> <end time>** cutting a time window

 - **zoom all X with X =1,2,3,..** zoom all traces by X

- **zoom <tracer> X** zoom a trace (tracer) by X
- **zoom all 1** un-zoom all traces
- **hide <tracer>** hide tracer
- **hide _distance(:113)** hide all the traces with distance bigger than 113
- **display h:all** displays all the traces
- **del <tracer>** delete traces, either all of them or 3-8
- **del _comp~(z)** delete all the N and E components
- Display more information
 - **time (exit via e)** information of the absolute and relative time
 - **am <start time> <end time>** information of the max. and min. amplitude
 - **am** the time window will be chosen by mouse
 - **markphases <tracer>** mark phases (requires distance and depth)
- Filtering
 - **Makef** creating a filter which needs following parameters:
 Filter type: bp
 FFt (f) or recursive ® filter: f
 Input type (f=Hz, t=sec, w=ang.freq.): t
 Corner 1: freq/time: 0.5
 Corner 2: freq/time:5.0
 Order of filter: 2
 Name of output file (no extension): bp0550s_o2
 - **fili f <filename>** load filter named <filename>
 - **filter f all** apply the filter on all traces
 - **doublefil <filename>** apply the filter named <filename> on all traces on both sides
- **CALC**

is used to carry out all arithmetic operations and saves the result in a variable.

 - **calc <type> &<var> = <operand1> [<operation> [<operand2>]]**
 - **calc t &dt = ^start(1) tdiff ^origin(1)** calculates the difference time between the origin time and the start time of the first trace. The result will be saved in the

variable dt.

- **calc r &test = 5 * "dt**
- **calc i &cnt = "cnt + 1**

○ CALL

the functions can be called, for example the function that calculates the theoretical travel time or the slowness of a phase.

- **call <subfunction> <p1> <p2> ... <pN>**
- **call slowness p ^distance(2) ^depth(2) &slo**
calculates the slowness of the P wave using the distance and depth of the second trace. The result will be saved in variable slo.

○ Printing

- **hc x1 p** saves the display window as a Postscript file (SH_HC001.ps) in the directory *~/shfiles/shscratch*